

title

Proposición 1. Sea $k \in \mathbb{N}$ y $f \in \mathcal{C}^k([-1/2, 1/2])$ 1-periódica

$$\implies \exists C_k < \infty : \forall N \in \mathbb{N} : \|S_N f - f\|_\infty \leq \frac{C_k}{N^{k-1/2}}.$$

Demostración: Por el `lem-serie-fourier-derivada`/lema 1, $\forall n \in \mathbb{Z} \setminus \{0\} : \widehat{f^{(k)}}(n) = (2\pi i n)^k \widehat{f}(n)$. Además, como $f \in \mathcal{C}^k([-1/2, 1/2]) \subset \mathcal{L}^2([-1/2, 1/2])$, por Plancherel, tenemos que

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} |f^{(k)}(x)|^2 dx = \sum_{n \in \mathbb{Z}} |\widehat{f^{(k)}}(n)|^2 = \sum_{n \in \mathbb{Z}} 4^k \pi^{2k} |n|^{2k} |\widehat{f}(n)|^2. \quad (1)$$

Por otro lado, por la `prop-criterio-dini`/proposición 1, $\sum_{n \in \mathbb{Z}} \widehat{f}(n) e^{2\pi i n x} = \lim_{N \rightarrow \infty} S_N f(x) = f(x)$. Entonces,

$$\begin{aligned} \|S_N f - f\|_\infty &= \left\| \sum_{|n| > N} \widehat{f}(n) e^{2\pi i n x} \right\|_\infty \leq \sum_{|n| > N} |\widehat{f}(n)| = \sum_{|n| > N} |\widehat{f}(n)| |n|^k \cdot \frac{1}{|n|^k} \\ &\stackrel{\text{C-S}}{\leq} \left(\sum_{|n| > N} |\widehat{f}(n)|^2 |n|^{2k} \right)^{1/2} \left(\sum_{|n| > N} \frac{1}{|n|^{2k}} \right)^{1/2} \end{aligned}$$

Ahora, usando la ecuación (1), tenemos que

$$\begin{aligned} \|S_N f - f\|_\infty &\leq \frac{1}{2^k \pi^k} \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} |f^{(k)}(x)|^2 dx \right)^{1/2} \left(2 \sum_{n=N+1}^{\infty} \frac{1}{n^{2k}} \right)^{1/2} \\ &\leq \frac{\sqrt{2}}{2^k \pi^k} \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} |f^{(k)}(x)|^2 dx \right)^{1/2} \left(\int_N^{\infty} \frac{1}{x^{2k}} dx \right)^{1/2} \\ &= \frac{\sqrt{2}}{2^k \pi^k} \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} |f^{(k)}(x)|^2 dx \right)^{1/2} \left(\frac{1}{(2k-1)N^{2k-1}} \right)^{1/2} = \frac{C_k}{N^{k-1/2}}, \end{aligned}$$

$$\text{donde } C_k = \frac{\sqrt{2}}{2^k \pi^k \sqrt{2k-1}} \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} |f^{(k)}(x)|^2 dx \right)^{1/2} < \infty. \quad \blacksquare$$